EXPERIMENT 3 – EXPLORATORY DATA ANALYSIS (EDA) – DATA CLEANING AND DATA TRANSFORMATION

Aim

To identify and handle missing values, duplicate records, outliers, and inconsistent data, and to perform data transformation techniques such as normalization and encoding using the Python Pandas library for effective data preprocessing.

Procedure

1. Import the required Python libraries such as Pandas, NumPy, and Scikit-learn.
2. Load the tourism_dataset.csv file into a Pandas DataFrame.
3. Check and handle missing values, duplicate records, and outliers in the dataset.
4. Perform data transformation using Normalization, Label Encoding, and One-Hot Encoding.
5. Display the cleaned dataset and save it as a new CSV file for further analysis

Program

Program 1: Identify Missing Values

print(df.isnull())

Program 2: Count Missing Values

print(df.isnull().sum())

OUTPUT:

```
[1]: import pandas as pd
     import numpy as np
```

```
[4]: df = pd.read_csv(r"C:\Users\kalap\Downloads\tourism_dataset.csv")
     print(df)
     df.head()
     df.info()
```

```
        Location Country    Category  Visitors  Rating    Revenue  \
0     kuBZRkVsAR   India      Nature    948853    1.32   84388.38
1     aHKUXhjzTo     USA  Historical    813627    2.01  802625.60
2     dlrdYtJFTA  Brazil      Nature    508673    1.42  338777.11
3     DxmlzdGkHK  Brazil  Historical    623329    1.09  295183.60
4     WJCCQlepnz  France    Cultural    124867    1.43  547893.24
...          ...     ...         ...       ...     ...        ...
5984  xAzwnVKAqz     USA       Urban    828137    1.97  132848.78
5985  IfKotyaJFC  France      Nature    276317    3.53  325183.96
5986  bPyubCWGgA   Egypt       Beach    809198    3.37  927336.50
5987  kkWIucpBnu   Egypt    Cultural    808303    2.52  115791.43
5988  gHXUrdticm  France    Cultural     40939    4.65  957026.85

     Accommodation_Available
0                        Yes
1                         No
2                        Yes
3                        Yes
4                         No
...                      ...
5984                      No
5985                     Yes
5986                      No
5987                     Yes
5988                     Yes
```

Program 3: Replace Missing Values

```
numeric_columns = df.select_dtypes(include=np.number).columns

for col in numeric_columns:
    df[col].fillna(df[col].mean(), inplace=True)

print(df)
```

Program 4: Find and Remove Duplicate Records

```
print(df.duplicated())
```

Program 5: Detect Outliers (Revenue)

```
Q1 = df["Revenue"].quantile(0.25)

Q3 = df["Revenue"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR

upper = Q3 + 1.5 * IQR

outliers = df[(df["Revenue"] < lower) |
              (df["Revenue"] > upper)]
print(outliers)
```

Program 6: Normalize Numerical Columns

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

df[["Visitors","Rating","Revenue"]] = scaler.fit_transform(
    df[["Visitors","Rating","Revenue"]]
)

print(df[["Visitors","Rating","Revenue"]])
```

OUTPUT:

```
Q1 = df["Revenue"].quantile(0.25)

Q3 = df["Revenue"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR

upper = Q3 + 1.5 * IQR

outliers = df[(df["Revenue"] < lower) |
              (df["Revenue"] > upper)]

print(outliers)
```

```
Empty DataFrame
Columns: [Location, Country, Category, Visitors, Rating, Revenue, Accommodation_Available]
Index: []
```

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

df[["Visitors","Rating","Revenue"]] = scaler.fit_transform(
    df[["Visitors","Rating","Revenue"]]
)

print(df[["Visitors","Rating","Revenue"]])
```

```
      Visitors  Rating   Revenue
0     0.948813  0.0800  0.083448
1     0.813435  0.2525  0.802423
2     0.508137  0.1050  0.338098
3     0.622922  0.0225  0.294460
4     0.123899  0.1075  0.547429
...        ...     ...       ...
5984  0.827961  0.2425  0.131958
5985  0.275519  0.6325  0.324491
5986  0.809001  0.5925  0.927262
5987  0.808105  0.3800  0.114884
```

Program 7: Label Encoding

```
from sklearn.preprocessing import LabelEncoder

encoder = LabelEncoder()

df["Country"] = encoder.fit_transform(df["Country"])

df["Category"] = encoder.fit_transform(df["Category"])

df["Accommodation_Available"] = encoder.fit_transform(df["Accommodation_Available"])

print(df.head())
```

Program 8: One-Hot Encoding

```
encoded = pd.get_dummies(
    df,
    columns=["Country","Category","Accommodation_Available"]
)

print(encoded.head())
```

OUTPUT:

```
[16]: encoded = pd.get_dummies(
          df,
          columns=["Country","Category","Accommodation_Available"]
      )

      print(encoded.head())
      df.to_csv("cleaned_tourism_dataset.csv", index=False)
```

```
     Location  Visitors  Rating   Revenue  Country_0  Country_1  Country_2  \
0  kuBZRkVsAR  0.948813  0.0800  0.083448      False      False      False   
1  aHKUXhjzTo  0.813435  0.2525  0.802423      False      False      False   
2  dlrdYtJFTA  0.508137  0.1050  0.338098      False       True      False   
3  DxmlzdGkHK  0.622922  0.0225  0.294460      False       True      False   
4  WJCCQlepnz  0.123899  0.1075  0.547429      False      False      False   

   Country_3  Country_4  Country_5  Country_6  Category_0  Category_1  \
0      False      False       True      False       False       False   
1      False      False      False       True       False       False   
2      False      False      False      False       False       False   
3      False      False      False      False       False       False   
4      False       True      False      False       False       False   

   Category_2  Category_3  Category_4  Category_5  Accommodation_Available_0  \
0       False       False        True       False                      False   
1       False        True       False       False                       True   
2       False       False        True       False                      False   
3       False        True       False       False                      False   
4        True       False       False       False                       True   

   Accommodation_Available_1  
0                       True  
1                      False  
2                       True  
3                       True  
4                      False  
```